

FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

7 Week: 3 April 2017

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P7.31 (a) 55.5 J (b) 64.5 J

P7.37 (a) 4.56 kJ (b) 4.56 kJ (c) 6.34 kN (d) 422 km/s² (e) 6.34 kN
(f) Explanation

P7.41 (a) +2.5 J (b) -9.8 J (c) -12 J

P7.43 (a) to (c) -196 J (d) explanation

P7.44 (a) It does not depend on the path (b) 35.0 J

P7.51 (a) 40.0 J (b) $\Delta U = -40.0$ J (c) 62.5 J

P7.61 (a) $\vec{F}_1 = (20.5\hat{i} + 14.3\hat{j})$ N; $\vec{F}_2 = (-36.4\hat{i} + 21.0\hat{j})$ N (b) $\sum \vec{F} = (-15.9\hat{i} + 35.3\hat{j})$ N (c) $\vec{a} = (-3.18\hat{i} + 7.07\hat{j})$ m/s² (d) $\vec{v}_f = (-5.54\hat{i} + 23.7\hat{j})$ m/s
(e) $\Delta \vec{r} = (-2.30\hat{i} + 39.3\hat{j})$ m (f) 1.48 kJ (g) 1.48 kJ (h) Explanation

P7.62 (a) $a = 4.01 \times 10^4$ N/m^{1.8} and $b = 1.80$ (b) 295 J

P7.63 $x = 0.131$ m

P13.30 (a) -4.77×10^9 J (b) 569 N (c) explanation

P13.33 (a) 1.84×10^9 kg/m³ (b) 3.27×10^6 m/s² (c) -2.08×10^{13} J

P13.35 (a) $U_{\text{total}} = 1.67 \times 10^{-14}$ J (b) explanation